
Katherine Piper

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Education

Carnegie Mellon University | Pittsburgh, Pennsylvania August 2024 – Present
Doctor of Philosophy | Computational Mechanics Est. April 2029
Master of Science | Computational Mechanics December 2025
California Institute of Technology | Pasadena, California September 2019 – June 2024
Bachelor of Science | Mechanical Engineering

Research Experience

Carnegie Mellon University | Pittsburgh, Pennsylvania August 2024 – Present
Graduate Researcher
- Constrained optimization machine learning methods for physics and engineering problems
DEVCOM Army Research Laboratory | Aberdeen, Maryland Summer 2026
Research Associate (Advisors: Jarek Knap and Kevin LoGrande)
- Advancing Diffusion Maps Surrogates with Constrained Optimization
NASA Jet Propulsion Laboratory | Pasadena, California
Academic Part Time Employee October 2023 – August 2024
Year Round Intern May 2023 – September 2023
- Additive Manufacturing Center (AMC) and Two-Phase Thermal Technology Laboratory
- Methods led to 100 times cost reduction and 42% performance improvement over state-of-the-art TPL design
- Two NASA New Technology Reports and US Patent submitted
California Institute of Technology | Pasadena, California
Summer Undergraduate Research Fellowship June 2022 – June 2024
Research Advisor – Vinamra Agrawal, Los Alamos National Laboratory
- Implementation of a model for ductile fracture in functionally graded additively manufactured metals

Awards

CEE Research Spotlight: 1 of 4 Featured Posters August 2026
CMU CEE Fall 2026 Poster Competition
Journeyman Fellowship June-August 2026
DEVCOM Army Research Laboratory
Grand Prize: Sustainable Impact Award – \$4,000 March 2026
Gresham Smith Innovation Hackathon
Graduate Fellowship 2024 - 2029
Carnegie Mellon University College of Engineering

Research Grants

ACCESS – EXPLORE – National Science Foundation Computing Grant September 2024 – Present

Publications and Patents

(Under Review) K. Piper and S. N. Roberts. Beyond Porosity and Permeability: An Alternative Design Space for Passive Capillary Evaporators. Submitted to *Joule*, September 2026.
(Provisional Patent) K. Piper, S. N. Roberts, and T. Daimaru. Novel Geometric Considerations for Solid and Porous Design in Loop Heat Pipe Evaporators. September 2025. *Assigned to California Institute of Technology*
(In Preparation) K. Piper, L. T. Biegler, N. Trask, and K. Dayal. Direct Nonlinear Optimization Enables Reduced Order Models for Contact.
K. Piper and V. Agrawal. Ductile fracture in functionally graded materials: Insight into crack behavior within the gradient interface. arXiv preprint arXiv:2411.18642 (2024)

Media Coverage and Public Engagement

Students pitch sustainability innovations at inaugural Hackathon March 4, 2026
Carnegie Mellon University Press Release - Project Featured as First Place Team
Loop Heat Pipes for Smallsat Swarms (LHPss) October 2024
Scott Roberts: NASA JPL FY24 Progress Report - Named Key Contributor